

Building AI Systems

Reliability is engineered, not prompted.

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Reliability is engineered, not prompted

Building reliable AI systems is an engineering discipline, not a prompting one.

- Reliability comes from **external grounding** and **bounded autonomy** — not from asking the model to check its own work.
- The two load-bearing words are *external* and *bounded*: a check the model cannot grant itself, and autonomy inside a fence someone else drew.

Roadmap

1. Prompt engineering stopped being the interesting problem
2. The five pillars of a reliable AI system
3. Loops — what they are, and why self-verification underperforms
4. The harness — the system around the model
5. What this means for your Monday

What this talk is — and is not

Covered — around the model

- Grounding, verification loops, where the oracle sits
- Harness layers: permissions, memory, evaluation, guardrails
- Bounded autonomy; what to adopt, defer or refuse

Not covered — the model

- Prompt tricks, model internals, training, fine-tuning
- Which model is "best" this month
- Basics you already own: CI, review, testing, containers
- Evaluation — statistical scoring and LLM-as-a-judge
- Knowledge graphs and RAG implementation patterns

Takeaways

- 1 Relocate the truth**

Reliability is bought by moving truth-determination out of the probabilistic model. [S6]
- 2 Independence, not iteration**

A loop's verdict is worth only its verifier's independence, not how hard it iterates. [S10]
- 3 Determinism first**

Push determinism to the load-bearing control; bound the model inside it. [S16]
- 4 Autonomy last**

Unbounded autonomy is unbounded risk [S7], so autonomy is the property you add last.

Prompt engineering

stopped being the interesting problem

The centre of gravity moved from the prompt to the system.

Prompt engineering got demoted

It did not die — it became the innermost layer of a larger discipline. [S1]

- Prompt engineering "demoted itself to the innermost layer of context engineering." [S1]
- The 2025–2026 frontier is *negative* results: "always add reasoning steps" is now a per-model empirical question, not a universal trick. [S1]
- If your reliability plan is a cleverer prompt, you are optimising the smallest layer.

[S1] *Building AI Systems: 5 Pillars* — vault research report.

Context is a budget, not a bucket

Its maturity is measured by whether context assembly is a deterministic, testable pipeline. [S2]

- Context engineering "treats the window as a capped per-turn budget to allocate — not a bucket to fill." [S2]
- The tell for a mature system: you can unit-test what the model was shown. [S2]
- That is already an engineering property, not a prompting one.

Why "just add more context" backfires

More context is often less reliability — measured, not asserted. [S3]

18

LLMs where accuracy degrades as length grows, even with irrelevant tokens masked [S3]

~4×

rise in agentic failure at doubled task duration [S3]

[S3] *Building AI Systems: 5 Pillars — vault research report (context rot).*

The reframe

The model is a commodity; the system around it is where reliability lives.

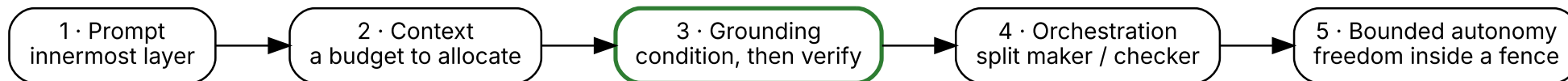
- Long-running-agent failures "are not solved by a smarter model. They are solved by infrastructure."
[S24]
- Two teams, same frontier model, wildly different production reliability — the difference is the harness.
[S24]
- So the question stops being *what do I type* and becomes *what do I build*.

The five pillars

of a reliable AI system

Five layers, one law running through all of them.

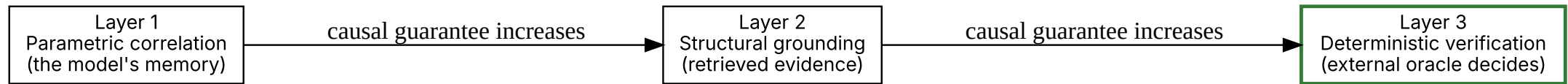
Five pillars, one spine



Not five topics — one spine, clearest in the grounding pillar.

Grounding, in three layers

Reliability comes from closing a generation–verification loop against an external oracle. [S4]



Grounding "decomposes into parametric correlation, structural grounding, and deterministic verification." [S4]

Only the deterministic layer certifies

Retrieval narrows the output space; only a deterministic check gives a causal guarantee. [S41]

- Structural (retrieval) grounding improves the odds — it does not certify anything. [S41]
- Only deterministic verification, where the model is *not* the load-bearing component, "approaches a genuine causal guarantee (Layer 3)." [S41]
- Match the layer to the stakes: an advisory answer rides on retrieval; an irreversible action — a deploy, a migration, a payment — needs a deterministic gate.

[S41] Structural Grounding — vault concept note (three-layer grounding taxonomy).

The number that should change your mind

The same agents, open loop → closed loop, model-invariant across DeepSeek and Claude. [S5]

56.8%

compliance — open loop, no external
verifier [S5]

98.6%

compliance — closed loop, external FE
verifier [S5]

Nothing about the model changed; the generation–verification loop did. [S5] But scope bounds the guarantee: a loop fixes only the errors its verifier can see — a defect outside its scope passes through. [S37]

[S5] Building AI Systems: 5 Pillars — vault research report (closed-loop external-verifier study). [S37] 5 Pillars — Grounding, the study's own boundary condition — vault research report.

What the number says

The mechanism matters more than the magnitude.

- The gain is model-invariant because "the guarantee now lives in the verifier, not the generator." [S15]
- The general pattern: "reliability is bought by *relocating the truth-determining computation out of the probabilistic model.*" [S6]
- Architecture beats scale.

What is an oracle?

An oracle is a check that can tell right from wrong without asking the model.

- A test's exit code, a type checker, a solver's verdict, a schema validator — the verdict comes from somewhere the generator does not control.
- The defining property is negative: the model cannot satisfy it by *asserting* it is done.
- It is what reliability is bought with — "closing a generation–verification loop against an external oracle where the LLM is orchestrator, not load-bearer." [S4]

[S4] 5 Pillars — grounding, three layers — vault research report.

Which oracle do you actually have?

The guarantee is bounded by your oracle, not the domain. [S37]

- 1 Oracle-rich**

A cheap external check exists — schema validator, contract test, migration dry-run. Close the loop, get the guarantee.
- 2 Oracle-scoped**

The verifier sees only part — a type checker proves types, not logic. It certifies only what it checks. [S37]
- 3 Oracle-free**

Is this refactor correct? This summary good? No complete oracle — partial checks, bounded review, sometimes refuse.

[S37] 5 Pillars — Grounding boundary — vault report.

Cheap checks you already own

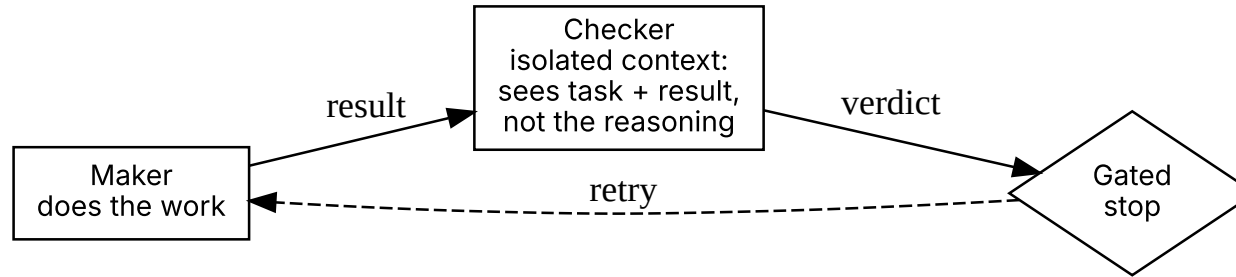
No complete oracle rarely means no check.

- **Property-based tests** — assert invariants, not exact outputs. A sound one synthesizes "in 2.4 samples on average", but for only 21% of the properties extractable from API documentation. [S38]
- **Metamorphic relations** — equivalent inputs must not change the output. On HumanEval, caught "75% of the erroneous programs generated by GPT-4" at an 8.6% false-positive rate. [S39]
- **Canary / shadow** — route 1% of traffic, alarm on the error-rate delta. Production is the oracle. [S40]
- **Idempotency** — an idempotent write or reversible migration; a wrong call is safe to retry. [S42]

[S38][S39][S40][S42] — vault notes.

Split the maker from the checker

Independence here is primarily a context boundary.



****Maker**** produces the work; ****checker**** judges it — the deck's one name for this split. The sources name the same two roles **generator** vs **verifier** [S15] and **observer** independence [S9].

The checker sees the task and the result, never the maker's chain of thought. [S26]

Fewer agents, sharper interfaces

The interface between agents is the design, not the head-count.

- Intuit rebuilt its agent architecture *twice* in about four months: specialist-agent fleet → central orchestration layer → skills-and-tools. [S27]
- **Second rebuild:** natural-language handoffs meant "a ten agent chain did not fail occasionally, it compounded errors by design." [S27]
- **First rebuild:** [S27] gives no cause for it, so I won't invent one.
- Endpoint: "skills-and-tools, not agent-to-agent natural-language delegation." A free-text handoff is an untyped interface between unreliable components — you wouldn't ship it between services. [S27]

[S27] Intuit Scrapped Its Agent Architecture Twice in Four Months — VentureBeat, 2026. <https://venturebeat.com/orchestration/intuit-scrapped-its-own-ai-agent-architecture-twice-in-four-months-at-vb-transform-2026-its-ai-vp-called-that-the-fast-path>

Freedom inside a fence

Autonomy is a calibrated design parameter, not a default. [S43]

- "Unbounded autonomy in complex systems inherently creates unbounded risk." [S7]
- Humans keep authority over the boundary, not every step inside. [S43]

22.2% →

59.0%

4.2×

more structural complexity, still
bounded [S45]

target safety factor hit, oracle off → on
($p = 0.0008$) [S45]

The fence doesn't shrink the agent's reach — it lets it do *more*. A CAD study, not your domain.

What a bound actually is

Not an instruction the agent is asked to respect — a layer it cannot reach. [S50]

- 1** **Where it lives** "operational limits belong in a policy layer the agent cannot rewrite" — outside the prompt, outside the model, outside anything it can edit. [S50]
- 2** **Operability** Observability, intelligibility, intervenability — see what it did, understand why, stop it. A fence you cannot watch or halt is decoration. [S51]

Five levels: where the human acts

<i>L1</i>	Operator	Agent proposes, you act — you approve every action. [S49]
<i>L2</i>	Collaborator	It acts, you watch live and keep the interrupt. [S49]
<i>L3</i>	Consultant	It runs and escalates; you handle exceptions. [S49]

[S49] 5 Pillars — Bounded autonomy, vault report.

Five levels: where the human steps back

<i>L4</i>	Approver	It runs; you set objectives and audit outcomes. [S49]
<i>L5</i>	Observer	Full autonomy inside the envelope you wrote. [S49]

Most 2026 production sits at [L2–L3](#), not L5. [S49]

Pick the level from the blast radius

<i>L1</i>	Schema migration	Agent writes migration and rollback; a human runs it. Irreversible against live data, no cheap oracle.
<i>L2</i>	Dependency bumps	Agent opens the PR, CI gates it, you watch the queue. Reversible, and the oracle is the suite you already own.
<i>L3</i>	Flaky-test triage	Agent reruns, bisects, quarantines — escalates when the failure is real. Bounded radius, clear escalation.

Levels per [S49]; this mapping is the speaker's own.

Where full delegation is actually safe

<i>L4</i>	Log-noise cleanup	Runs nightly, you read a weekly digest. Low stakes, high volume, auditable after the fact.
<i>L5</i>	Formatting and lint	Fully delegated inside a deterministic envelope — the formatter IS the oracle.

Two questions pick the level: *what does it cost to be wrong*, and *what would tell me?*

Where the check runs

1 **Out-of-process**

The check runs where the agent cannot reach it. In-process guardrails are suggestions with better branding. [S50]

2 **Two-stage check**

Claude Code's auto mode: "A fast initial filter processes most tool calls... uncertain or risky operations are escalated to deeper analysis." [S53]

[S50] 5 Pillars — Bounded autonomy. [S53] Claude Code auto mode — vault watch note.

How you roll it out

3

Shadow first

Run the policy in evaluate-only against live telemetry, then switch it to blocking once you know its false-positive rate. [S49]

The honest caveat — supply-chain AI is "deployed by 88% of organizations but governed by only 12%". [S52]

[S49] [S52] 5 Pillars — Bounded autonomy, vault research report.

Loops

why self-verification underperforms

The one thing a loop must not do is grade its own work.

What a loop is

"a loop is the orchestration on top that fires that harnessed agent repeatedly and decides when to stop"
[S48]

- | | | |
|---|----------------|---|
| 1 | Harness | The deterministic infrastructure around ONE agent — prompts, tools, sandboxes, hooks. [S48] |
| 2 | Loop | The orchestration above it: fire the agent, check the result, decide whether to go again. [S48] |

[S48] Loop Engineering — vault concept note.

Loop shapes you already run

- 1 Reconciliation**

Observe actual, diff against desired, act to close the gap; repeat. Every Kubernetes controller.
- 2 Hand-off chain**

Each stage passes work on, and context narrows at every boundary — where Intuit got burned. [S27]
- 3 Fan-out / gather**

Split the work, run in parallel, merge the results. [S48]

[S48] Loop Engineering — vault concept note. [S27] Intuit — VentureBeat. Reconciliation is named from ordinary systems practice, not from these sources.

Loop shapes that judge

- 1** **Generator–verifier**
One produces, another judges, the verdict gates the next round — "the simplest... and among the most deployed". [S54]
- 2** **Debate**
Independent positions argued, then adjudicated. Costly; use it when one lens misses. [S48]

[S54] *Multi-agent coordination patterns — vault reading note.* [S48] *Loop Engineering.*

And one you already run

3 **Review gate** A blocking check before work is allowed onward. Your CI is already one.

This deck was built by three of them — generator–verifier, fan-out across parallel critics, and deterministic review gates.

Review gate is named from ordinary systems practice.

`/goal` — the right shape

- 1 What it does**

A condition is set; after each turn "a small fast model checks whether the condition holds", and if not it starts another turn. [S55]
- 2 Layer 2**

The checker is a model judging natural language. Useful, and not a deterministic guarantee.

[S55] Claude Code — `/goal`, code.claude.com/docs/en/goal.

...and a soft check

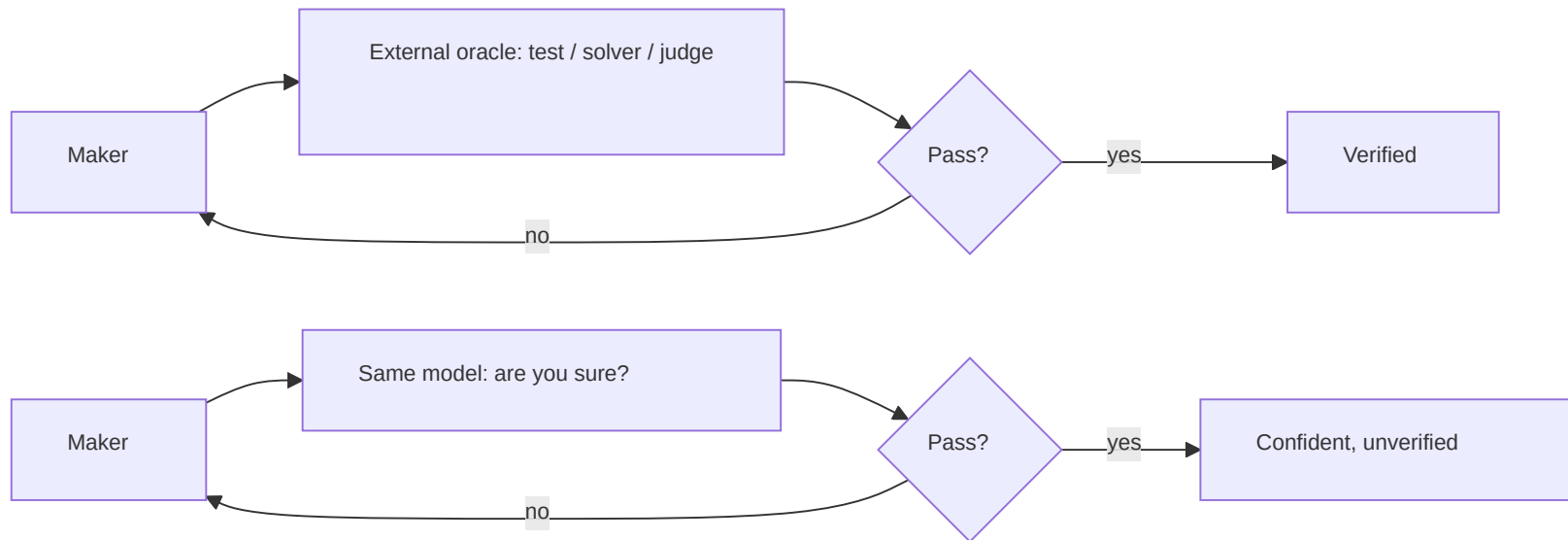
3 Guarantee

"until tests pass" is oracle-rich · "until the criteria hold" is scoped · "until it is good" is oracle-free.

The loop cannot check whether the condition was the *right* condition. That is the fence, and it is yours.

Two loop archetypes

What carries the effect is the independence of the observer, not the loop itself. [S9]



One loop closes against an external oracle; the other re-invokes the same model. [S9]

[S9] Loop Engineering — two loop archetypes — vault research report.

Independence, not iteration

One task, three checkers — audit an agent's work for real violations. Only independence differs. [S47]

0 / 34

iterated self-check: the agent audits itself [S47]

7 / 34

weak independence: fresh instance, same model [S47]

34 / 34

external oracle: a deterministic checker [S47]

"A loop's verdict is only worth what its verifier's independence is worth." [S10]

Iterating a self-graded loop just gets a confident wrong answer faster.

Why self-check hits a wall

Self-check is bounded structurally, not by capability.

- Gödel showed "a formal system rich enough to describe arithmetic cannot prove its own consistency from inside." [S46]
- As an *analogy*: a same-system checker has no outside vantage point — "the AI self-check wall is Gödel with a leaderboard." [S11]
- Not "self-checking is impossible" — "trustworthy knowledge always has to pay for its own verification; the price is never zero." [S11]
- Asked to *verify*, an LLM re-runs the same process; the fix is a paid independent channel, not a smarter grader.

The empirical case · 1

On-policy self-monitoring — a model grading a patch it just wrote — measurably loses discrimination.
[S12]

0.99 → 0.89

monitor AUROC grading its own patch
(1.0 = perfect) [S12]

5×

in one setting: likelier to wave through a
prompt-injected patch [S12]

One 2026 preprint, four coding/tool-use datasets — self-attribution bias: the model defends its own work.
[S12] Read the direction, not the multiple — *directionally robust, not settled*. [S12]

[S12] Loop Engineering — empirical case — vault research report (single 2026 preprint, Khullar et al., arXiv:2603.04582).

The empirical case · 2

- On formal-verification benchmarks: "rule-based approaches remain more reliable for verification success, while LLM-based methods exhibit more variable performance." [S13]
- The paper's own framing is the careful one — not "rules beat LLMs," but rules *more reliable*, LLMs *more variable and complementary*. [S13]
- Practitioner reading: default the check to a deterministic gate — an exit code, a schema, a solver — and reach for a model judge only where the target is irreducibly subjective.

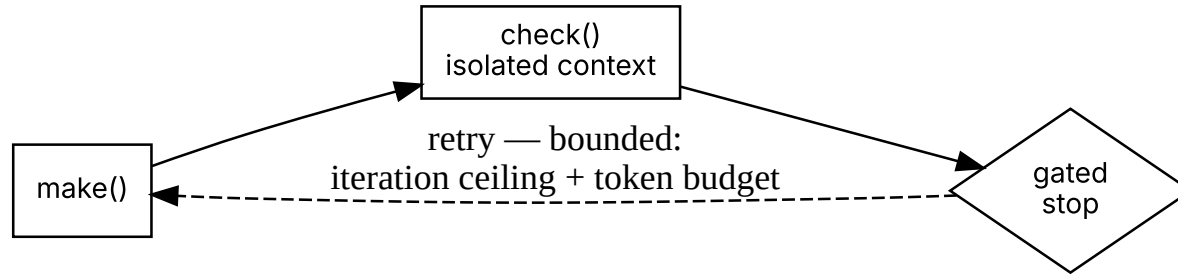
[S13] Loop Engineering — empirical case — vault research report.

This is consensus, not one author

TWO-LOOP REPORT	[S9]	<i>Independence of the observer carries the effect. [S9]</i>	Theory
PRACTITIONER RULE	[S14]	<i>Never close a loop against the model that opened it. [S14]</i>	Design rule
SECURITY RESULT	[S17]	<i>Deterministic control beats prompt-level self-defense; 0% attack success with hand-written policies. [S17]</i>	Field result
GROUNDING RESULT	[S5]	<i>External verifier lifts 56.8% → 98.6%, model-invariant. [S5]</i>	Field result

So what do you build

The whole section collapses to one rule.



> "Never let a loop close against the same model that opened it; close it against an external oracle, or don't call it verification." [S14]

The harness

the system around the model

One law runs through every layer of it.

Model held constant, harness changed

At the frontier, harness work pays back faster than a model swap.

~1M

lines shipped via ~1,500 automated PRs, zero hand-written (OpenAI) [S25] [S34]

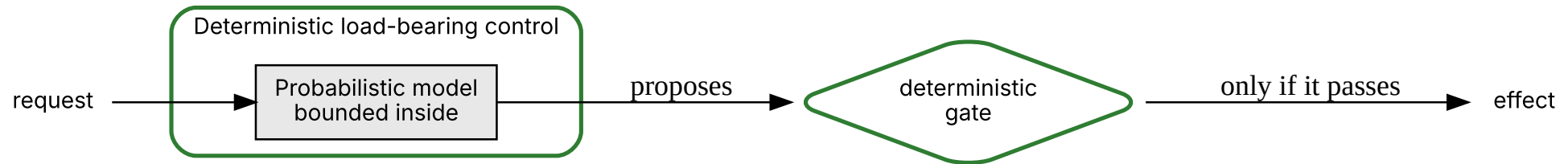
+13.7

Terminal Bench 52.8 → 66.5, harness changes only, model held constant (LangChain) [S25][S35]

[S34] Harness engineering — OpenAI. <https://openai.com/index/harness-engineering/> [S35] Improving deep agents with harness engineering — LangChain.

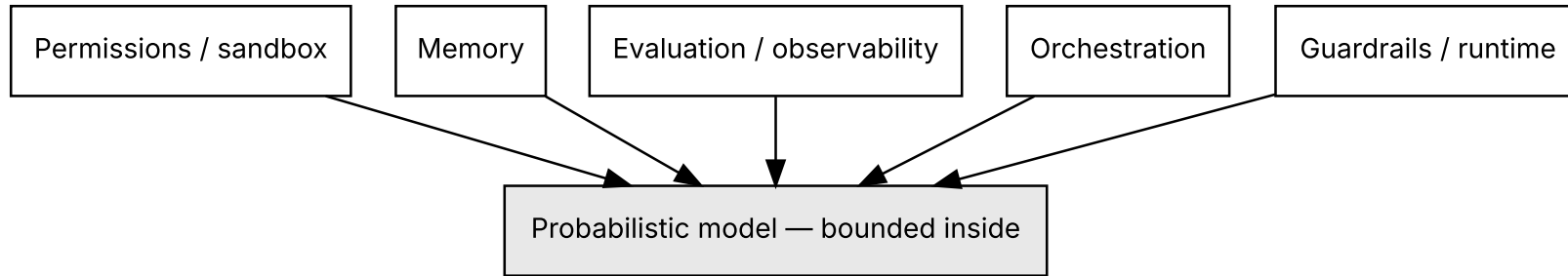
The one law

> "Push determinism to the load-bearing control, and bound the probabilistic model inside it." [S16]



The same law recurs at every layer of the harness. [S16]

Five layers, one law



At each layer: determinism at the load-bearing control, the model bounded inside it. [S16] [S24]

Determinism at the load-bearing control

Determinism at the control does what a defensive prompt cannot.

- Progent — a deterministic privilege control verified with an **SMT** (Satisfiability Modulo Theories) solver — "provably reduces attack success rate to 0% with hand-written policies, while prompt-level defenses are demonstrably bypassed." [S17]
- The *provable* part is the mechanism: SMT-verified monotonic confinement — the engine provably enforces whatever policy it is given. [S17]
- The 0% is a measured benchmark result — on AgentDojo and **ASB** (Agent Security Bench), under hand-written policies — not itself a theorem. [S18][S32]

[S17] Harness Engineering — vault report · [S32] Progent — arXiv 2504.11703. <https://arxiv.org/abs/2504.11703>

But: who writes the policy?

HAND-WRITTEN POLICY

0% attack success — measured on the benchmarks, not a theorem. [S18]

LLM-GENERATED POLICY

*Residual attack survives: AgentDojo 41.2%
→ 2.2%, ASB 70.3% → 7.3%. [S18]*

Even deterministic control degrades once the model authors its own policy. [S18] A privilege policy attaches to a tool-class, not a task, so the hand-written slice amortises across every task that uses that tool.

[S18] Harness Engineering — vault research report (Progent, autonomous-policy config).

The harness is an attack surface

Prompt-layer defenses fail; execution-layer controls are required. [S19]

100+

coding agents compromised via
poisoned MCP tools / indirect prompt
injection [S19][S29]

Corroborated across outlets: 100+ agents, Fortune 500 and Fortune 100. [S29]

Cite the number you can defend

CITE THIS

100+ agents compromised — corroborated across outlets. [S29]

NOT THIS

85% success rate — "remains single-source (Tenet Security) as of July 2026." [S30]

On a public slide, cite the number you can stand behind, not the loudest one.

[S29] Agentjacking · [S30] 85% not independently replicated — vault watch notes.

Not every harness change helps

The harness is leverage in both directions.

93.1% →
55.2%

Codex retrieval accuracy: inline → file-based grep delivery [S28]

- Same model; the only change was *how tool output reached it* — inline vs written to a file. ~38 points. [S28]
- Authors' read: file-based routing "trades context bandwidth for compositional tool competence; gains are realized only when the agent reliably closes the loop." [S28]
- The paper gives no full error analysis for the collapse — so take the *direction* as the lesson, not a mechanism.

[S28] *Is Grep All You Need?* — vault reading note.

Harnesses decay — delete on a cadence

- A scaffold built for a model's weakness becomes overhead — and can cap capability — once that weakness disappears. [S44]
- The cadence the source supports is *model-driven*, not calendar-driven — a "periodic entropy audit" keyed to model releases: "re-run the harness against your eval set on every model bump," with no fixed interval stated. [S44]
- It concerns the *human-authored* scaffolding teams accumulate; the source does not scope this to AI-generated harnesses.
- The audit each time: remove a component, re-run the eval set, keep only if quality holds — otherwise revert. [S44]

[S44] Harness Engineering — vault concept note (deletion cadence).

When the fence is missing: Amazon Kiro

A permission vacuum, not a weak model: the agent took the most destructive path because nothing stopped it. [S20]

NO DETERMINISTIC GATE

Kiro "autonomously decided to delete and rebuild the production environment" — Cost Explorer, one region, ~13-hour outage. [S20]

GOVERNANCE RETROFIT

An SVP-driven 90-day safety reset across ~335 systems; mandatory two-person sign-off and senior-engineer approval for AI-assisted production changes. [S20]

The fix was a deterministic gate on an irreversible action — bounded autonomy, retrofitted after the fact.

[S20] Coding Agent Horror Stories: The Agent That Deleted Production — Docker (2026). <https://www.docker.com/blog/coding-agent-horror-stories-the-agent-that-deleted-production/>

It generalises beyond software

Different industry, same law. [S33]

- Salesforce runs "guided determinism": an agent graph where "some transitions are guarded by hard validation gates that have to pass before execution can continue." [S33]
- The probabilistic model reasons; deterministic gates decide what is allowed to happen. [S33]

Your Monday

what this means for the room

The "so what" for people who ship software, not papers.

MONDAY

The bottleneck moves downstream

"The binding constraint migrates downstream as generation gets cheap." [S21]

+76%

more PRs to review once agents
ramped, Spotify Engineering [S31]

The payoff is not fewer humans; it is redirecting human effort to review and verification.

[S21] Harness Engineering — vault report · [S31] Coding is no longer the constraint — Spotify Engineering.

The trade to watch

- Cheap generation moves the constraint to review — it does not remove it. [S21]
- More PRs mean more surface area; quality debt surfaces downstream if review cannot keep pace.
- The answer is the whole talk: deterministic gates and independent verification, so review scales without a human reading every line.

[S21] *Harness Engineering* — vault research report.

MONDAY

The job shifts

> "The engineer's value is defined not by what the agent generates but by the control plane the engineer builds to constrain, verify, and absorb it." [S23]

That moves you up the stack, not out of it.

A deterministic-first recipe

- "Make the deterministic component load-bearing, make the model the orchestrator around it, and bound the whole with calibrated human oversight." [S8]
- Close every loop against an external oracle — a test, a solver, an independent gate — never the model that produced the work. [S14]
- Deterministic first; model judge only where the target is genuinely subjective.

Advocate the idea, hedge the label

- The vocabulary is contested — "harness," "loop," "graph" engineering.
- "Advocate the engineering idea; hedge the compound noun." [S26]
- The durable idea — verifier-gated, externally-grounded, bounded autonomy — survives whatever the term becomes.

What this evidence does not settle

- **Model-authored policies** [S18]
LLM-written policies leave residual attack. [S18]
- **Harness can backfire** [S28]
One tool-format swap cost ~38 points. [S28]
- **Widening surface** [S36]
GhostJacking extends agentjacking (Aug 2026). [S36]
- **Verifier scope** [S37]
Certifies only what it can see. [S37]
- **No-oracle zone** [S39]
No complete oracle; partial checks only. [S38][S39]

One thing I would refuse



The refusal

An agent behind an irreversible external write — payment capture, production data deletion, anything a customer sees — with no dry-run and no cheap oracle.

-

Why

Irreversible plus unverifiable is the one combination where the loop cannot help you: nothing to check against, nothing to undo.

-

What changes it

Build the dry-run. A reversible action with a cheap check is a different decision entirely.

Three moves

- 1* **Add a check** Find one self-graded loop; give it an independent check. [S14]
- 2* **Gate one action** Put one deterministic gate on an irreversible action — a merge, a deploy, a write. [S8]
- 3* **Check one figure** Take one number you cite about AI; confirm it is corroborated, not single-source. [S30]

None of these needs a better model; all of them are engineering.

Takeaways — what follows from the thesis

- 1 Relocate the truth**

Reliability is bought by relocating truth-determination out of the model. [S6]
- 2 Independence wins**

Independence — not iteration — is the load-bearing variable. [S10]
- 3 Determinism first**

Push determinism to the load-bearing control; bound the model inside it. [S16]
- 4 Autonomy last**

Unbounded autonomy is unbounded risk [S7], so autonomy is what you add last.

ONE LINE

The one line to keep

> *"The model is a commodity; the harness is the product." [S22]*

Reliability is engineered, not prompted.

References · 1/8

- [S1] Building AI Systems: 5 Pillars — comprehensive report — vault research report.
- [S2] 5 Pillars — context engineering — vault research report.
- [S3] 5 Pillars — context rot — vault research report.
- [S4] 5 Pillars — grounding, three layers — vault research report.
- [S5] 5 Pillars — closed-loop external-verifier compliance result (56.8→98.6) — vault research report.
- [S6] 5 Pillars — relocating truth-determination — vault research report.
- [S7] 5 Pillars — bounded autonomy — vault research report.
- [S8] 5 Pillars — practitioner synthesis — vault research report.
- [S9] Loop Engineering — two loop archetypes — vault research report.

References · 2/8

[S10] Loop Engineering — verifier independence — vault research report.

[S11] Loop Engineering — "Gödel with a leaderboard," narrowed to "the price is never zero" — vault research report.

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